Informal Sector Enterprise Surveys Raw Database:
Background Information about the Cross-City Harmonized Data

Introduction

The Informal Sector Enterprise Surveys are products of the Enterprise Analysis Unit that measures the characteristics and activities of unregistered businesses. These informal businesses are everywhere in the world, though they are almost always missing from official records, listings of active businesses, and business-level surveys. The Informal Sector Enterprise Surveys are rigorous surveys that use an adaptive, geographically based sampling method to fill this data gap. This document provides additional information on the survey methodology, questionnaire design and construction of the Informal Sector Enterprise Surveys raw database.

By presenting broadly comparable data at the level of informal businesses based on the representative samples at the city level, the Informal Sector Enterprise Surveys fill a large gap in the data availability and provide a database to better understand industry characteristics and practices of a segment of the economy that remains largely unexplored.

The rest of this note is organized as follows. Section I provides a short summary of survey methodology. The questionnaire design and composition are discussed in Section II. Section III briefly outlines how sampling weights are calculated. Section IV describes the construction of the harmonized cross-city raw database. The final section explains certain practicalities such as the identifier variables that are included in the dataset.

Section I: Survey Methodology

A challenge to conducting a representative survey of informal sector businesses is the lack of a proper sampling frame since these businesses are not registered and, therefore, they are almost always absent from official registries or any other potential sampling frame. The data are collected through face-to-face interviews of owners and managers following a two-stage sampling process. The first stage uses a methodology called Adaptive Cluster Sampling (ACS). ACS requires a well-defined geographical area, which for the Informal Sector Enterprise Surveys is typically an urban center. Often, the boundaries of an urban center match a set of administrative boundaries. Nevertheless, when appropriate, a wider (possibly outside of administrative boundaries) or narrower urban area is used to best capture the extent of informal business activity in an area. A grid of evenly sized squares, called Block Areas or BAs, is then overlaid on a map of this delineated area.

An initial sample of BAs is selected at random (often within strata) without replacement. Within an enumerated BA, some basic information is listed on all encountered informal businesses, including through observation for those that refuse the exercise or those that are unavailable at the time of fieldwork. ACS is ‘adaptive’ in the sense that, using a pre-defined threshold number of informal businesses, all neighboring BAs of a BA that meets the threshold are subsequently sampled until there are no BAs that meet this expansion requirement.[1] This process allows for the first-stage weights to be calculated, enabling the Informal Sector Enterprise Surveys to be geographically representative of the urban areas where it is implemented. For further details on this methodology, see Aga et al., 2022,

For increased precision, the Informal Sector Enterprise Surveys uses the stratified version of ACS. After the spatial grid is created, BAs in the grid of each urban center are categorized into five strata: residential, commercial/industrial, mixed (commercial and residential), market center, and open areas.[2] The stratification is typically based on different local resources available. Note that blocks stratified as market centers are not selected using ACS but rather through simple random sampling (SRS).

The second stage of the Informal Sector Enterprise Surveys involves the random selection of informal businesses within a BA to participate in an interview that lasts approximately 20–25 minutes. This interview consists of a standardized and tested questionnaire, properly designed for the circumstances of interviewing informal businesses. The selection for these longer interviews is done in real time and the fieldwork team does not have discretion on how businesses are selected. As a result, a second-stage weight can be applied.

Through this two-stage sampling process, the interviewed sample is representative of the informal businesses that are located in the urban center covered by the Informal Sector Enterprise Surveys and population indicators can be estimated through applying sampling weights.

Section II: Questionnaire design and composition

Surveying informal businesses requires a well-developed instrument to be used for the interview. The questionnaire used in the implementation of the Informal Sector Enterprise Surveys is the result of a long process of experimentation. The World Bank Enterprise Analysis unit has explored various approaches of surveying informal businesses around the world for several years. The process involved consultations with statistical offices across several countries, evaluation of the questionnaires that have previously been used to survey informal businesses, and application of the extensive experience of the team which implements the World Bank Enterprise Surveys (WBES) of formal firms. This trial-and-error learning process has led to the following principles when designing the appropriate questionnaire.

Informal businesses should not be interviewed using questionnaires designed for registered businesses. The primary reason for this is that terminology and definitions used in such questionnaires reflect experiences of legal, registered entities, and may not be suited for informal businesses. Accounting and technical terms commonly used in formal business surveys, such as 'equity', 'income statement', or capacity utilization' must be avoided. It is important to recognize that informal businesses operate in ways that are fundamentally different from formal businesses. Therefore, not only the terms used in the questionnaire, but also concepts need to be adapted to

[1] Different patterns of expansion can be implemented through ACS. The Informal Sector Enterprise Surveys typically activate all surrounding BAs of those meeting the expansion threshold. If the expansion process is considerably longer than expected, a Team Task Leader (TTL) can force a stop to expansions.
[2] There is a sixth category for blocks that are physically inaccessible. This category is excluded from the sampling.

ensure that pertinent measures are being collected. For example, when measuring labor inputs, it is important to capture non-remunerated labor or support from family, so survey questions must be worded accordingly and/or added.

The interviewees of the Informal Sector Enterprise Surveys tend to have lower levels of education than the typical respondent of a formal business survey, and some may even be unable to read or write. Consequently, direct yes/no questions are preferable, and commonly used aids for interviewing, such as 'show cards', should be avoided. The language used as well as the question structure should be simple, straightforward, and able to be read quickly. This means that questions must be formulated in ways that require little or no explanation. Questions that require great cognitive effort, such as those involving counterfactuals or vignettes scenarios, are avoided.

To maximize the effectiveness of the questionnaire, questions requiring mathematical computations or complex numerical responses should be minimized, and those requiring responses in percentages should be avoided. Importantly, since many informal businesses may not keep any written accounts of their activities, it is not advisable to have questions that use accounting terms or timeframes, such as ‘fiscal year’. Questions that use annual time horizons are also not advisable as many informal businesses may have a short time-horizon. Instead, to minimize recall bias, that is likely strong in the absence of written records, shorter periods of reference such as last month, are preferable.

Importantly, surveying a segment of the population that operates on the border of legality carries challenges with sensitivities. Non-response is not uncommon. To ensure the participation of respondents, it is important to go beyond convincing them that their responses will be anonymized. In particular, it is important to provide assurances that the information will not be used for prosecution. Questions should be formulated in ways that are the least invasive to respondents. Given these challenges of implementation the team that implements the survey must be thoroughly trained to administer the questionnaire in the form of a conversation. This not only increases the participation of informal businesses in the survey, but also avoids respondent reticence. The need to approximate a normal conversation during the survey interview further emphasizes the importance of simplicity throughout the questionnaire.

The Informal Sector Enterprise Surveys questionnaire covers a range of socio-economic topics that not only provide a granular picture of informal businesses and their owners, but also discern salient patterns and features of a typical informal business. While the exact set of questions asked during interviews changes across countries, a core set of questions has been consistently asked, enabling cross-country analysis. Topics include the demographics background of the owners, the origin of the informal businesses, their operations, sales and employment, sources of financing, business practices as well as the topic of informality itself. For more information about the questionnaire, see Aberra et al. (2022).

Section III: Sampling Weights

To estimate population parameters, weights are applied to survey samples. In survey designs following standard SRS, the selection probability of all units is known before the actual data collection. Hence, weights can be derived as the inverse of the probability of selection. A similar process is followed for ACS, with the probability of selection (within stratum) adjusted to account for the adaptive selection process.

Let n_h denote the total number of starting blocks (PSUs) selected by stratum h, with $h = 1, \ldots, H$ indexing each stratum. Let N_h denote the total number of blocks in the universe of stratum h. Within stratum, starting PSUs are selected using SRS without replacement. Whenever the number of informal businesses in a given starting block is above a pre-defined threshold, determined by city, all surrounding blocks are enumerated as long as they are not inaccessible or market centers (market centers are selected by SRS and are not eligible for expansion). The enumeration of surrounding blocks continues until no block meets the pre-defined threshold requirement. This process produces a set of *networks*, which are formed by a group of contiguous blocks that all meet the expansion threshold. Thompson (1990, 1991) defines a network as any group of blocks where the enumeration of one block would lead to the enumeration of all blocks in the network. This definition helps show how, within a network, the probability of selection can be estimated. Specifically, let $m_{h,i}$ denote the total number of blocks of stratum h in a network i.[3] In the case that a starting block i does not meet the threshold, it is considered as a network with a size of $m_{h,i} = 1$. Then, the inclusion probability of a block in network i is defined as π_i and given by[4]:

$$\pi_i = 1 - \prod_{h=1}^{H} \left[\frac{\binom{N_h - m_{h,i}}{n_h}}{\binom{N_h}{n_h}} \right]$$

The inverse of π_i provides the first-stage weight for each block i. For market center blocks which are completed following SRS, the weight is calculated as $\frac{n_h}{N_h}$.

In providing population estimates using ACS (with stratification), Thompson (1990, 1991) has shown that these weights produce unbiased estimators; however, in many cases there will be informal businesses that are enumerated in blocks that are not part of the starting selection and do not meet the expansion threshold. These blocks are called *edge units*, and the unbiased estimators call for omitting these blocks. The reason for this is that for edge units, there are neighboring, surrounding blocks that have not been enumerated, and so the actual probability of selection for edge units cannot be known. Due to the expense of fieldwork and the fact that there are often interviews obtained in edge units, the choice has been made to calculate weights for edge units as well. These weights are estimated according to the formula for π_i with the following substitutions: the term $m_{h,i}$ is replaced with $(2m_{h,i} + e_h)$ where e_h is a dummy that equals 1 for edge units of strata h and 0 otherwise, and n_h is replaced with $2n_h$.[5] The resulting weights have a known but directionally agnostic bias on population estimates when edge units are retained in this manner. As

[3] Note that within a network, all blocks are treated as the same, and so some ACS literature indexes by network rather than block.

[4] If there are no blocks with stratum h in the network i, then N_h and n_h are set to 0. Note that with networks of size $m_i = 1$, all blocks will be within one stratum and the probability of selection simplifies to the same probability of the initial SRS selection (within the stratum).

[5] In the special scenario where an edge unit is of a stratum that the adjacent network does not contain, the formula uses N_h and n_h of stratum h (i.e., avoiding adjustment detailed in footnote 4 above).

of the publication of the 2022 Informal Sector Enterprise Surveys in cities in India, we feel it is worth the bias to retain the information collected.

The published Informal Sector Enterprise Surveys datasets consist of all interviews, respondents of which are selected randomly within the enumerated blocks. These datasets therefore include weights for each interview that are the result of multiplying the first-stage weight (probability of selection of a BA) with the second stage weight (probability of selection of a business within selected BAs). Second-stage weights are given by the inverse of the ratio of the number of interviews completed in the block and the total number of informal businesses found in the block. Since within blocks, there typically are both refusals to the enumeration exercise and informal businesses that cannot be reached (and are recorded by observation), assumptions about the eligibility of these businesses may be needed. Different assumptions generate different adjustment factors. Specifically, three versions of the second-stage weights are provided based on different assumptions that are used in estimating the total number of informal businesses found in a block, with varying ways of handling refusals and unavailable businesses, as follows:

Assumption	**Variable Name**	**Condition of Inclusion**
Strict	wstrict	All confirmed informal businesses only
Median	wmedian	All confirmed informal businesses and refusals that **don't** have signage/permits on display
Weak	wweak	All confirmed informal businesses and all refusals

The strict assumption calculates the total number of informal businesses found in a block by including only businesses that are confirmed to be informal. The median assumption assumes that businesses that refused to participate in the survey and do not have signage and permits on display are part of the informal sector of the economy; these businesses are added to those confirmed to be informal. The weak assumption treats all refusals, regardless of signage and permits, as part of the informal economy. Therefore, by definition:

$$wweak \geq wmedian \geq wstrict$$

Data users can choose to apply whichever weight they conclude is appropriate under these assumptions. All indicators and analysis conducted by the Informal Sector Enterprise Surveys team use the *wmedian* weights.

Data users should note that there is a debate on whether and how to use weights in regressions (see Deaton, 1997, pp.67; Solon et al. 2013; Lohr, 1999, chapter 11, Cochran, 1977, pp.150). There is no strong, large-sample econometric argument in favor of using weighted estimation for a common population coefficient if the underlying model varies per stratum (stratum-specific coefficient): both simple OLS and weighted OLS are inconsistent under regular conditions. However, weighted OLS has the advantage of providing an estimate that is independent of the sample design. More generally, if the regressions are descriptive of the population, then weights should be used. If the models are developed as structural relationships or behavioral models that may vary for different parts of the population, then there is no reason to use weights.

Section IV: Database Construction

The Informal Sector Enterprise Surveys raw database combines the data collected under the methodology described in Section I, using the questionnaire described in Section II, across all surveyed cities and countries. In the process of putting this cross-country data together, special care was given to comparing the questionnaires used across the surveys and properly mapping variables to ensure compatibility. Only the questions that were worded the same way or with minor changes were put together. For example, informal businesses in Zimbabwe were asked, "In the last completed month, did this business or activity experience harassment by government officials or police?" In subsequent surveys, the reference period was changed from "in the last completed month" to "over the last three months." In such a scenario, responses from Zimbabwean businesses cannot be included among those from other countries, as the reference period of the question significantly changed for meaningful comparability. If the response categories differed across the surveys, efforts were made to accurately map the categories into common groups. If this was not possible, the questions were treated as different and were not mapped together. While there is no common questionnaire available in the standard format, the data users can rely on the variable labels to get a general idea of the question and consult the documentation file (which includes the questionnaire) of the corresponding country-level survey for the exact wording of each question.

Section V: Practical Aspects of Using the Data

To link the harmonized data with the Informal Sector Enterprise Surveys indicators database or with the city-level datasets that are available on the web portal, we recommend using the variable *idstd*.[6] This is a unique identifier of each interview in the entire set of the Informal Sector Enterprise Surveys interviews. It corresponds to the same variable that is included in all relevant the Informal Sector Enterprise Surveys datasets. Most variables in the dataset are numeric, except those with suffix “x”, which are string variables. The variables *wweak*, *wmedian*, *wstrict* are the sampling weights (consolidated from the two stages) corresponding to the different weighting criteria (see Section III for details). The sampling weights enable inferences to the population of informal businesses in each city and are strongly recommended to be used in data analysis. The variable *strata* indicates the sampling design stratum within each city. The variable *id_square* identifies businesses that have been interviewed in the same block. The variable *id_cluster* indicates the cluster each block belongs to.

The harmonized cross-city raw dataset is current as of the date noted in the heading of this document. It will be updated upon the publication of additional Informal Sector Enterprise Surveys datasets. All changes will be documented in detail.

We ask users of the data to cite the dataset as follows:

> World Bank Group, Enterprise Analysis Unit. 2022. “Informal Sector Enterprise Surveys”. www.enterprisesurveys.org/

[6] The data portal is available at http://www.enterprisesurveys.org/

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